aici aveti exercitii rezolvate din examenele finale date de Ghiu la CD pe parcursul anilor. langa unele exercitii am scris "poza x" si se refera la examenul din care am luat exercitiul. pozele le gasiti in "examene Ghiu CD" (doar subiectele pentru final, complete, nerezolvate)

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EXERCIȚII EXAMEN

POZA 3

① $f: \mathbb{R}^3 \to \mathbb{R}$, $f(x,y,z) = x^3 + 5y^3 - 3x^2y + yz^2 - 12y - 6z$

pct. de extrem?

$\frac{\partial f}{\partial x} = 3x^2 - 6xy$

$\frac{\partial f}{\partial y} = 15y^2 - 3x^2 + z^2 - 12$

$\frac{\partial f}{\partial z} = 2yz - 6$

$\Rightarrow \begin{cases} x^2 - 2xy = 0 \Rightarrow x(x-2y) = 0 \\ 15y^2 - 3x^2 + z^2 - 12 = 0 \\ yz - 3 = 0 \Rightarrow z = \frac{3}{y} \end{cases}$

I $x = 0$

$15y^2 + \frac{9}{y^2} - 12 = 0$

$15y^4 - 12y^2 + 9 = 0 \quad | :3$

not. $y^2 = t > 0$

$5t^2 - 4t + 3 = 0$

$\Delta = 16 - 60 < 0 \Rightarrow$ pct. $x = 0$, $y \notin \mathbb{R}$

II $x = 2y \Rightarrow y = \frac{x}{2} \Rightarrow z = \frac{6}{x}$

$15y^2 - 3(2y)^2 + \left(\frac{3}{y}\right)^2 - 12 = 0$

$15y^2 - 12y^2 + \frac{9}{y^2} - 12 = 0$

$3y^2 + \frac{9}{y^2} - 12 = 0$

$3y^4 - 12y^2 + 9 = 0 \quad | :3$

not. $y^2 = t > 0$

$t^2 - 4t + 3 = 0$

$t^2 - 3t - t + 3 = 0$

$t(t-1) - 3(t-1) = 0$

$(t-1)(t-3) = 0$

1. $t_1 = 1 \Rightarrow y_1 = -1, x_1 = -2, z_1 = -3 \Rightarrow A(-2,-1,-3)$

$y_2 = 1, x_2 = 2, z_2 = 3 \Rightarrow B(2,1,3)$

2. $t_2 = 3 \Rightarrow y_3 = -\sqrt{3}, x_3 = -2\sqrt{3}, z_3 = -\sqrt{3} \Rightarrow C(-2\sqrt{3}, -\sqrt{3}, -\sqrt{3})$

$y_4 = \sqrt{3}, x_4 = 2\sqrt{3}, z_4 = \sqrt{3} \Rightarrow D(2\sqrt{3}, \sqrt{3}, \sqrt{3})$

} puncte critice

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$\frac{\partial^2 f}{\partial x^2} = 6x - 6y \qquad \frac{\partial^2 f}{\partial x \partial y} = -6x$

$\frac{\partial^2 f}{\partial y^2} = 30y \qquad \frac{\partial^2 f}{\partial x \partial z} = 0$

$\frac{\partial^2 f}{\partial z^2} = 2y \qquad \frac{\partial^2 f}{\partial y \partial z} = 2z$

$$H_f(x,y,z) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial z \partial x} & \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z^2} \end{pmatrix} = \begin{pmatrix} 6x-6y & -6x & 0 \\ -6x & 30y & 2z \\ 0 & 2z & 2y \end{pmatrix}$$

$\Delta_1 = 6(x-y)$

$\Delta_2 = 180y(x-y) - 36x^2 = -180y^2 + 180xy - 36x^2 = 36(-5y^2 + 5xy - x^2)$

$\Delta_3 = (x-y)(360y^2 - 24z^2) - 72x^2y$

1. $A(-2,-1,-3)$

$\left.\begin{array}{l} \Delta_1 < 0 \\ \Delta_2 > 0 \\ \Delta_3 > 0 \end{array}\right\} \Rightarrow$ A nu este pct. de extrem

2. $B(2,1,3)$

$\left.\begin{array}{l} \Delta_1 > 0 \\ \Delta_2 > 0 \\ \Delta_3 < 0 \end{array}\right\} \Rightarrow$ B nu este pct. de extrem

3. $C(-2\sqrt{3}, -\sqrt{3}, -\sqrt{3})$

$\left.\begin{array}{l} \Delta_1 < 0 \\ \Delta_2 > 0 \\ \Delta_3 < 0 \end{array}\right\}$ C este pct. de MAXIM $\Rightarrow f(-2\sqrt{3}, -\sqrt{3}, -\sqrt{3}) = 12\sqrt{3}$

4. $D(2\sqrt{3}, \sqrt{3}, \sqrt{3})$

$\left.\begin{array}{l} \Delta_1 > 0 \\ \Delta_2 > 0 \\ \Delta_3 > 0 \end{array}\right\}$ D este pct. de MINIM $\Rightarrow f(2\sqrt{3}, \sqrt{3}, \sqrt{3}) = -12\sqrt{3}$

$\min_{(x,y,z) \in \mathbb{R}^3} f(x,y,z) = -12\sqrt{3}$

$\max_{(x,y,z) \in \mathbb{R}^3} f(x,y,z) = 12\sqrt{3}$

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POZA 4

② $K = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 12x-16y-75\}$, $f: K \to \mathbb{R}$, $f(x,y) = x^2+y^2$. mim, max = ?

$x^2-12x+y^2+16y \le -75$

observăm ec. unui cerc și vrem să formăm binoame

$x^2-12x+36+y^2+16y+64 \le -75+36+64$

$(x-6)^2+(y+8)^2 \le 25$

centrul $(6,-8)$, raza 25

I. $\text{Int } K = \{(x,y) \in \mathbb{R}^2 \mid (x-6)^2+(y+8)^2 < 25\}$

$\frac{\partial f}{\partial x} = 2x$

$\frac{\partial f}{\partial y} = 2y$

$\Rightarrow \begin{cases} 2x=0 \\ 2y=0 \end{cases} \Rightarrow x=0, y=0 \Rightarrow O(0,0) \notin K$

$\Rightarrow$ nu există pct. critice pe disc



II. Fr K

PARAMETRIZAREA CERCULUI $x = C_x + R\cos\alpha$, $y = C_y + R\sin\alpha$

$x = 6+5\cos\alpha$

$y = -8+5\sin\alpha$

$f(x,y) = (6+5\cos\alpha)^2 + (-8+5\sin\alpha)^2 =$

$= 36+60\cos\alpha+25\cos^2\alpha+64-80\sin\alpha+25\sin^2\alpha =$

$= 100+60\cos\alpha-80\sin\alpha+25(\cos^2\alpha+\sin^2\alpha) =$

$= 125+60\cos\alpha-80\sin\alpha$

$f(x,y) = g(\alpha) = 125+60\cos\alpha-80\sin\alpha, \ \alpha \in [0,2\pi]$

$g'(\alpha) = -60\sin\alpha-80\cos\alpha$

II. Fr K METODA MULTIPLICATORILOR LAGRANGE

fie $g(x,y) = (x-6)^2+(y+8)^2-25$

$\boxed{L(x,y,\lambda) = f(x,y)+\lambda g(x,y)}$

$L(x,y,\lambda) = x^2+y^2+\lambda[(x-6)^2+(y+8)^2-25]$

$\frac{\partial L}{\partial x} = 2x+2\lambda(x-6)$

$\frac{\partial L}{\partial y} = 2y+2\lambda(y+8)$

$\frac{\partial L}{\partial \lambda} = (x-6)^2+(y+8)^2-25$

$\Rightarrow \begin{cases} 2x+2\lambda(x-6)=0 \\ 2y+2\lambda(y+8)=0 \\ (x-6)^2+(y+8)^2-25=0 \end{cases}$

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- $x + \lambda(x-6) = 0$
  $x + \lambda x - 6\lambda = 0$
  $x(1+\lambda) - 6\lambda = 0$
  $x = \frac{6\lambda}{1+\lambda}$

- $y + \lambda(y+8) = 0$
  $y + \lambda y + 8\lambda = 0$
  $y(1+\lambda) + 8\lambda = 0$
  $y = -\frac{8\lambda}{1+\lambda}$

- $\left(\frac{6\lambda}{1+\lambda} - 6\right)^2 + \left(-\frac{8\lambda}{1+\lambda} + 8\right)^2 - 25 = 0$
  $\left(\frac{6\lambda - 6 - 6\lambda}{1+\lambda}\right)^2 + \left(\frac{-8\lambda + 8 + 8\lambda}{1+\lambda}\right)^2 - 25 = 0$
  $\frac{36}{(1+\lambda)^2} + \frac{64}{(1+\lambda)^2} - 25 = 0$
  $\frac{100 - 25(1+\lambda)^2}{(1+\lambda)^2} = 0, \quad \lambda \neq -1$
  $100 - 25 - 50\lambda - 25\lambda^2 = 0$
  $-25\lambda^2 - 50\lambda + 75 = 0 \quad / \cdot \left(-\frac{1}{25}\right)$
  $\lambda^2 + 2\lambda - 3 = 0$
  $\lambda^2 + 3\lambda - \lambda - 3 = 0$
  $\lambda(\lambda - 1) + 3(\lambda - 1) = 0$
  $(\lambda - 1)(\lambda + 3) = 0$
  $\lambda_1 = 1 \Rightarrow x = 3, y = -4 \Rightarrow A(3, -4)$
  $\lambda_2 = -3 \Rightarrow x = 9, y = -12 \Rightarrow B(9, -12)$
  verif. dacă $A, B \in \mathcal{C}$
  $(3-6)^2 + (8-4)^2 = 25 \Rightarrow 9 + 16 = 25$ „A"
  $(9-6)^2 + (8-12)^2 = 25 \Rightarrow 9 + 16 = 25$ „A"
  $\Rightarrow A, B$ puncte de extrem $\in \mathcal{F} \setminus D$
  $f(3, -4) = 25$
  $f(9, -12) = 225$
  $\min_{(x,y) \in K} f(x, y) = 25$
  $\max_{(x,y) \in K} f(x, y) = 225$

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POZA 5

③ $f: A \to \mathbb{R}$, $f(x,y) = x^2 + y^2 + 4y + 1$

$A = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 6x + 4y - 12\}$

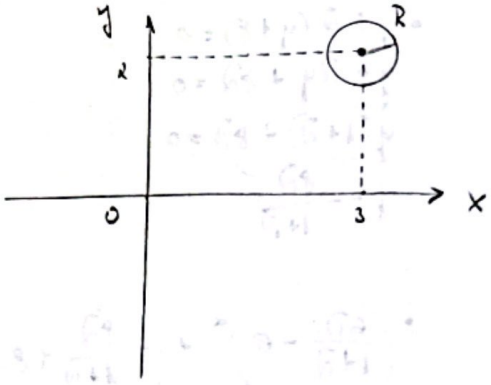
$\text{Int } A = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 < 6x + 4y - 12\}$

$x^2 + y^2 - 6x - 4y + 12 < 0$

$x^2 - 6x + 9 + y^2 - 4y + 4 - 1 < 0$

$(x-3)^2 + (y-2)^2 < 1$

discul de centrul $(3,2)$ și rază 1



$\frac{\partial f}{\partial x} = 2x$

$\frac{\partial f}{\partial y} = 2y + 4$ $\Rightarrow \begin{cases} 2x = 0 \\ 2y + 4 = 0 \end{cases} \Rightarrow x = 0, y = -2 \notin A$

$\Rightarrow \nexists$ puncte critice în $\text{Int } A$

$\text{Fr } A$ → folosim METODA MULTIPLICATORILOR LUI LAGRANGE

fie $L(x,y,\lambda) = f(x,y) + \lambda g(x,y)$, unde $g(x,y) = (x-3)^2 + (y-4)^2 - 1$

$L(x,y,\lambda) = x^2 + y^2 + 4y + 1 + \lambda[(x-3)^2 + (y-2)^2 - 1]$

$\frac{\partial L}{\partial x} = 2x + 2\lambda(x-3)$

$\frac{\partial L}{\partial y} = 2y + 4 + 2\lambda(y-2)$ $\Rightarrow \begin{cases} 2x + 2\lambda(x-3) = 0 \\ 2y + 4 + 2\lambda(y-2) = 0 \\ (x-3)^2 + (y-2)^2 - 1 = 0 \end{cases}$

$\frac{\partial L}{\partial \lambda} = (x-3)^2 + (y-2)^2 - 1$

$x + \lambda(x-3) = 0$

$x + \lambda x - 3\lambda = 0$

$x(1+\lambda) - 3\lambda = 0$

$x = \frac{3\lambda}{1+\lambda}$

$y + 2 + \lambda(y-2) = 0$

$y + 2 + \lambda y - 2\lambda = 0$

$y(1+\lambda) + 2 - 2\lambda = 0$

$y = \frac{-2(1-\lambda)}{1+\lambda}$

$\left(\frac{3\lambda}{1+\lambda} - 3\right)^2 + \left(\frac{-2(1-\lambda)}{1+\lambda} - 4\right)^2 - 1 = 0$

$\left(\frac{3\lambda - 3 - 3\lambda}{1+\lambda}\right)^2 + \left(\frac{2\lambda - 2 - 2 - 2\lambda}{1+\lambda}\right)^2 - 1 = 0$

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$$\frac{9}{(1+\lambda)^2} + \frac{(-6\lambda+6)^2}{(1+\lambda)^2} - 1 = 0$$

$$\frac{25+1-2\lambda-\lambda^2}{(1+\lambda)^2} = 0 \quad ; \quad \lambda \neq -1$$

$$\lambda^2 + 2\lambda - 24 = 0$$

$$\Delta = 4 + 4 \cdot 24 = 100$$

$$\lambda_{1,2} = \frac{-2 \pm 10}{2}$$

$\lambda_1 = 4 \Rightarrow x_1 = \frac{12}{5}, y_1 = \frac{6}{5} \in A\left(\frac{12}{5}, \frac{6}{5}\right)$.

$\lambda_2 = -6 \Rightarrow x_2 = \frac{18}{5}, y_2 = \frac{14}{5} \in A$

$$f\left(\frac{12}{5}, \frac{6}{5}\right) = 30$$

$$f\left(\frac{18}{5}, \frac{14}{5}\right) = 50$$

$$\Rightarrow \min_{(x,y) \in A} f(x,y) = 30 \; ; \; \max_{(x,y) \in A} f(x,y) = 50$$

④ $\int_0^{\frac{\pi}{3}} \sin^2 x \sqrt[3]{\operatorname{tg}^2 x}\, dx = I$ POZA - 1

$$\operatorname{tg} x = \frac{\sin x}{\cos x}$$

$$I = \int_0^{\pi/2} \sin^2 x \sqrt[3]{\frac{\sin^2 x}{\cos^2 x}}\, dx = \int_0^{\pi/2} \sin^2 x \cdot \frac{\sin^{2/3} x}{\cos^{2/3} x}\, dx =$$

$$= \int_0^{\pi/2} \sin^{\frac{8}{3}} x \cdot \cos^{-\frac{2}{3}} x\, dx$$

mot. $\sin^2 x = t \Rightarrow 2 \sin x \cos x\, dx = dt \Rightarrow \sin 2x\, dx = dt$

$x = 0 \Rightarrow t = 0$

$x = \frac{\pi}{2} \Rightarrow t = 1$

- $\sin^2 x = t \Rightarrow \sin x = \sqrt{t} \Rightarrow (\sin x)^{\frac{8}{3}} = (\sqrt{t})^{\frac{8}{3}} = t^{\frac{8}{6}} = t^{\frac{4}{3}}$
- $\sin^2 x + \cos^2 x = 1 \Rightarrow \cos x = \sqrt{1 - \sin^2 x} = \sqrt{1-t} \Rightarrow$
$\Rightarrow (\cos x)^{-\frac{2}{3}} = (\sqrt{1-t})^{-\frac{2}{3}} = (1-t)^{-\frac{1}{3}}$
- $dx = \frac{1}{2 \sin x \cos x}\, dt = \frac{1}{2\sqrt{t} \cdot \sqrt{1-t}} = \frac{1}{2\sqrt{t-t^2}}$

$I = \int$

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$$I = \int_0^1 t^{\frac{4}{3}} \cdot (1-t)^{-\frac{1}{3}} \cdot \frac{1}{2\sqrt{t-t^2}}\, dt =$$

$$= \int_0^1 t^{\frac{4}{3}} \cdot (1-t)^{-\frac{1}{3}} \cdot \frac{1}{2} \cdot [t(1-t)]^{-\frac{1}{2}}\, dt =$$

$$= \frac{1}{2}\int_0^1 t^{\frac{4}{3}-\frac{1}{2}} \cdot (1-t)^{-\frac{1}{3}-\frac{1}{2}}\, dt =$$

$$= \frac{1}{2}\int_0^1 t^{\frac{5}{6}} \cdot (1-t)^{-\frac{5}{6}}\, dt$$

$p-1 = \frac{5}{6} \Rightarrow p = \frac{11}{6}$

$q-1 = -\frac{5}{6} \Rightarrow q = \frac{1}{6}$

$$I = B\left(\frac{11}{6}, \frac{1}{6}\right) = \frac{\Gamma\left(\frac{11}{6}\right)\Gamma\left(\frac{1}{6}\right)}{\Gamma\left(\frac{11}{6}+\frac{1}{6}\right)} = \frac{\frac{5}{6}\Gamma\left(\frac{5}{6}\right)\cdot\Gamma\left(\frac{1}{6}\right)}{\Gamma(2)} = \frac{5}{12}\Gamma\left(\frac{1}{6}\right)\Gamma\left(1-\frac{1}{6}\right)$$

$$= \frac{5}{12}\cdot\frac{\pi}{\sin\frac{\pi}{6}} = \frac{5}{12}\cdot\frac{\pi}{\frac{1}{2}} = \frac{5\pi}{6}$$

✓ E BINE !

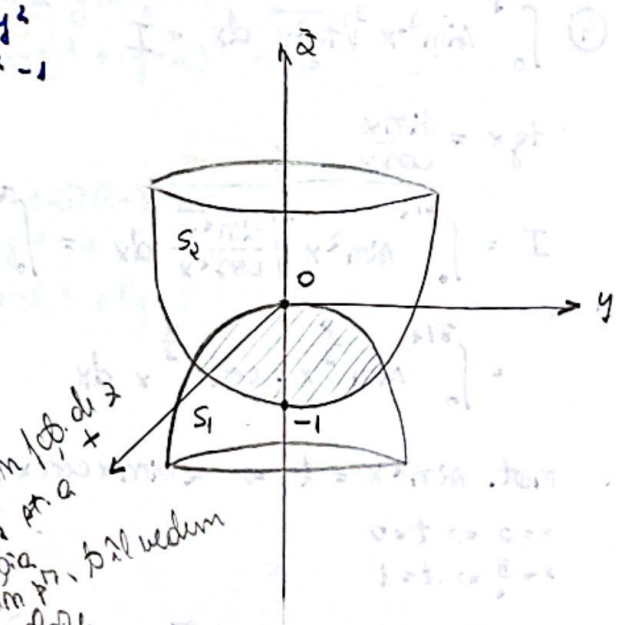
POZA 1

⑤ Sisd. vol. mărginit de $S_1: z = -x^2 - y^2$

$S_2: z = x^2 + y^2 - 1$

$S_1: x^2 + y^2 = -z \Rightarrow$ paraboloid
axă de simetrie Oz
deschidere în jos
secțiune cerc
vârf în origine

$S_2: x^2 + y^2 = z + 1 \Rightarrow$ paraboloid
axă de simetrie Oz
deschidere în sus
secțiune cerc
vârf $(0,0,-1)$



- scriem capetele în funcție de z
- egalăm S_1 cu S_2 pt a det. proiecția
- nu ne pasă ce dim pt., știl vedem
- coord. polare

$$V = \iiint_\Omega dx\,dy\,dz = \iint_L \left(\int_{x^2+y^2-1}^{-x^2-y^2} dz\right) dx\,dy =$$

$$= \iint_L (-x^2 - y^2 - x^2 - y^2 + 1)\, dx\,dy = \iint_L (-2x^2 - 2y^2 + 1)\, dx\,dy$$

$-x^2 - y^2 = x^2 + y^2 - 1$

$2x^2 + 2y^2 = 1 \quad /:2$

$x^2 + y^2 = \left(\frac{1}{\sqrt{2}}\right)^2$

cerc de centrul $(0,0)$ și rază $\frac{1}{\sqrt{2}} \Rightarrow L = pr_{xOy}\Omega = \left\{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le \left(\frac{1}{\sqrt{2}}\right)^2\right\}$

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trecem la coord. polare:

$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases}$, $r \in [0, \frac{1}{\sqrt{2}}]$, $t \in [0, 2\pi]$

$J(x,y) = r$

$$V = \int_0^{2\pi} \left(\int_0^{\frac{1}{\sqrt{2}}} \left(-2(r\cos t)^2 - 2(r\sin t)^2 + 1\right) r\, dr \right) dt =$$

$$= \int_0^{2\pi} \left(\int_0^{\frac{1}{\sqrt{2}}} \left(-2r^2 \underbrace{(\cos^2 t + \sin^2 t)}_{=1} + 1\right) r\, dr \right) dt =$$

$$= \int_0^{2\pi} \left(\int_0^{\frac{1}{\sqrt{2}}} (-2r^3 + r)\, dr \right) dt =$$

$$= \int_0^{2\pi} \left(-\frac{r^4}{2} + \frac{r^2}{2} \right) \Big|_0^{\frac{1}{\sqrt{2}}} dt =$$

$$= \int_0^{2\pi} \left(-\frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)^4 + \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)^2 \right) dt =$$

$$= \int_0^{2\pi} \left(-\frac{1}{2}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{1}{2} \right) dt =$$

$$= \left(-\frac{1}{8} + \frac{1}{4} \right) \cdot t \Big|_0^{2\pi} =$$

$$= \frac{1}{8} 2\pi =$$

$$= \frac{\pi}{4}$$ E BINE :)

POZA 3

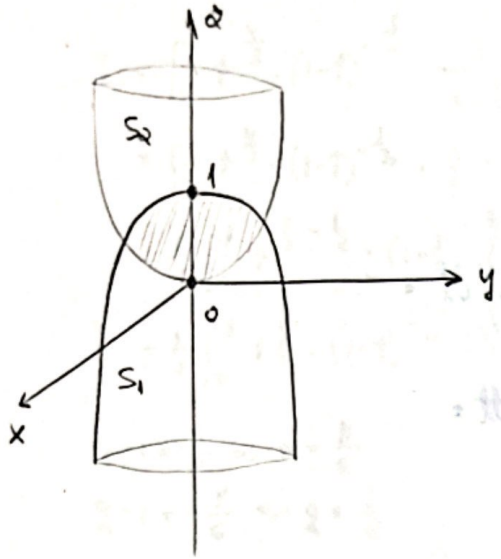
⑥ Std. vol. mărginit de $S_1: 1 - z = x^2 + y^2$

$S_2: x^2 + y^2 - z = 0$

$S_1: x^2 + y^2 = -(z-1)$ => paraboloid
- axă de sim. Oz
- deschidere în jos
- secțiune cerc
- vârf (0,0,+1)

$S_2: x^2 + y^2 = z$ => paraboloid
- axă de sim. Oz
- deschidere în sus
- secțiune cerc
- vârf în origine

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$$V = \iiint_\Omega dx\,dy\,dz = \iint_L \left(\int_{x^2+y^2}^{-x^2-y^2+1} dz \right) dx\,dy =$$

$$= \iint_L \left(-x^2 - y^2 + 1 - x^2 - y^2 \right) dx\,dy =$$

$$= \iint_L (-2x^2 - 2y^2 + 1)\, dx\,dy$$

$-x^2 - y^2 + 1 = x^2 + y^2$

$2x^2 + 2y^2 = 1$

$x^2 + y^2 = \left(\frac{1}{\sqrt{2}}\right)^2 \Rightarrow L = pr_{xOy}\,\Omega = \left\{ (x,y) \in \mathbb{R}^2 \,/\, x^2 + y^2 \le \left(\frac{1}{\sqrt{2}}\right)^2 \right\}$

ANALOG EX. 5 $\Rightarrow V = \frac{\pi}{4}$

POZA 4

④ Sad. vol. mult. mărginită de suprafețele: $S_1: \frac{x^2}{8} + y^2 = 3z$

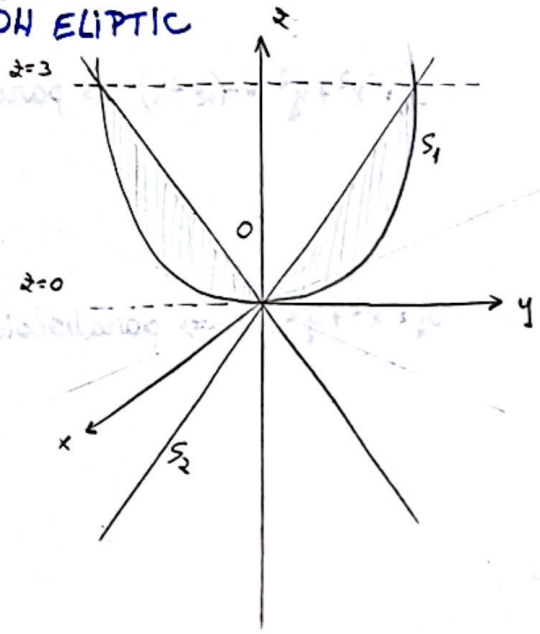
$S_2: \frac{x^2}{8} + y^2 = z^2$

$S_1: \frac{x^2}{8} + y^2 = 3z \;/:3 \Rightarrow \frac{x^2}{24} + \frac{y^2}{3} = z \Rightarrow$ PARABOLOID ELIPTIC

- axă de simetrie Oz
- deschidere în sus
- vârf în origine
- semiaxe $a = \sqrt{24}$, $b = \sqrt{3}$ ale secțiunii elipsă

$S_2: \frac{x^2}{8} + y^2 = z^2 \Rightarrow \frac{x^2}{8} + y^2 - z^2 = 0 \Rightarrow$ CON ELIPTIC

- conul se deschide pe Oz
- centru de simetrie în origine
- semiaxa $a = \sqrt{8}$ a secțiunii elipsă



$$V = \iiint_\Omega dx\,dy\,dz = \iint_D \left(\int_{\frac{x^2}{24}+\frac{y^2}{3}}^{\sqrt{\frac{x^2}{8}+y^2}} dz \right) dx\,dy =$$

$$= \iint_D \left(\sqrt{\frac{x^2}{8} + y^2} - \frac{1}{3}\left(\frac{x^2}{8} + y^2\right) \right) dx\,dy =$$

$$\begin{cases} \frac{x^2}{8}+y^2=3z \\ \frac{x^2}{8}+y^2=z^2 \end{cases} \Rightarrow z(z+3)=0 \Rightarrow \begin{matrix} z_1=0 \\ z_2=-3 \end{matrix}$$ am determinat z

- acotăm δ și punem la capete
- egalăm $S_1 \cap S_2$ și aflăm z
- înlocuim z în una din cele 2 și aflăm D
- coord. polare

înlocuim în ec: $\frac{x^2}{8}+y^2=3 \Rightarrow \frac{x^2}{72}+\frac{y^2}{9}=1$ elipsă

$$D=\left\{(x,y)\in\mathbb{R}^2 \,/\, \frac{x^2}{(6\sqrt{2})^2}+\frac{y^2}{3^2}\le 1\right\}$$

trecem la coord. polare pentru elipsă:

$$\begin{cases} x=a\cdot r\cos t \\ y=b\cdot r\sin t \end{cases}, r\in[0,1], t\in[0,2\pi], a=6\sqrt{2}, b=3$$

$J=a\cdot b\cdot r$

$$V=\int_0^{2\pi}\left(\int_0^1\left(\sqrt{\frac{(6\sqrt{2}\cdot r\cdot\cos t)^2}{8}+\frac{(3\cdot r\cdot\sin t)^2}{1}}-\frac{1}{3}\left(\frac{(6\sqrt{2}\cdot r\cdot\cos t)^2}{8}+\frac{(3\cdot r\cdot\sin t)^2}{1}\right)\right)\cdot 6\sqrt{2}\cdot 3\cdot r\,dr\right)dt =$$

$$=\int_0^{2\pi}\left(\int_0^1\left(\sqrt{\frac{72r^2\cos^2 t}{8}+9r^2\sin^2 t}-\frac{1}{3}\left(\frac{72r^2\cos^2 t}{8}+9r^2\sin^2 t\right)\right)\cdot 18\sqrt{2}\,r\,dr\right)dt =$$

$$=\int_0^{2\pi}\left(\int_0^1\left(\sqrt{9r^2(\cos^2 t+\sin^2 t)}-\frac{1}{3}\cdot 9r^2(\cos^2 t+\sin^2 t)\right)\cdot 18\sqrt{2}\,r\,dr\right)dt =$$

$$=\int_0^{2\pi}\left(\int_0^1(3r-3r^2)18\sqrt{2}\,r\,dr\right)dt =$$

$$=\int_0^{2\pi}\left(\int_0^1\left(3\cdot 18\sqrt{2}\,r^2-3\cdot 18\sqrt{2}\,r^3\right)dr\right)dt =$$

$$=\int_0^{2\pi}3\cdot 18\sqrt{2}\left(\frac{r^3}{3}-\frac{r^4}{4}\right)\Big|_0^1 dt =$$

$$=\int_0^{2\pi}3\cdot 18\sqrt{2}\cdot\frac{1}{12}\,dt$$

$$=\frac{9\sqrt{2}}{2}t\Big|_0^{2\pi}$$

$$=9\pi\sqrt{2}$$

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(7) Sa se det. vol. mult. $\left\{(x,y,z)\in\mathbb{R}^3 \,\middle|\, \frac{x^2}{64}+y^2+z^2\le 1,\ \frac{x^2}{64}+y^2\le\frac{(z+1)^2}{4}\right\}$

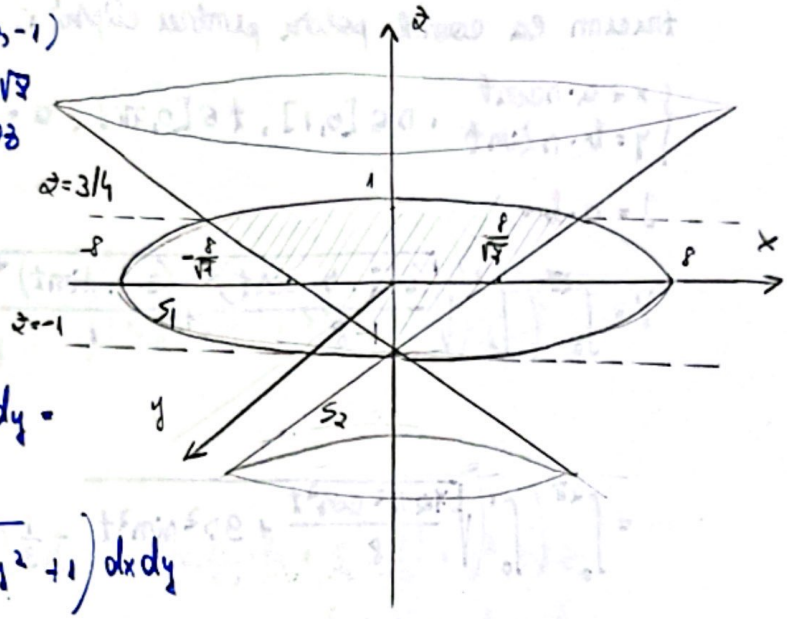
POZA 5

$S_1: \frac{x^2}{64}+y^2+z^2\le 1$ => interiorul unui ELIPSOID

- centrul de simetrie în origine
- semiaxele $a=8,\ b=1,\ c=1$

$S_2: \frac{x^2}{64}+y^2-\frac{(z+1)^2}{4}\le 0$ => interiorul unui CON ELIPTIC

- centrul de simetrie $(0,0,-1)$
- semiaxe $a=8,\ b=1,\ c=\sqrt{4}$
- conul se deschide pe Oz



$$V=\iiint_\Omega dx\,dy\,dz=$$

$$=\iint_D\left(\int_{\sqrt{\frac{4x^2}{64}+4y^2-1}}^{\sqrt{\frac{x^2}{64}+y^2+1}} dz\right)dx\,dy=$$

$$=\iint_D\left(\sqrt{\frac{x^2}{64}+y^2+1}-\sqrt{\frac{4x^2}{64}+4y^2+1}\right)dx\,dy$$

aflăm z' dim S_1 și S_2:

$$1-z^2=\frac{(z+1)^2}{4}$$

$$4(1-z)(1+z)=(z+1)^2$$

$$(z+1)(4(1-z)-(z+1))=0$$

$$(z+1)(6-5z)=0$$

$z_1=-1$
$z_2=6/5$ => dreptele care delimitează intersecția conurilor

înlocuim z în ec. a elipsei D:

$$\frac{x^2}{64}+y^2+\left(\frac{3}{4}\right)^2\le 1$$

$$\frac{x^2}{64}+y^2=\frac{4}{16}\ /\cdot\frac{16}{4}$$

$$\frac{x^2}{(\sqrt{28})^2}+\frac{y^2}{\left(\sqrt{\frac{4}{16}}\right)^2}=1$$ => ec. elipsei ce indică domeniul max.

$$\Rightarrow D=pr_{xOy}\Omega=\left\{(x,y)\in\mathbb{R}^2 \,\middle|\, \frac{x^2}{(\sqrt{28})^2}+\frac{y^2}{\left(\sqrt{\frac{4}{16}}\right)^2}\le 1\right\}$$

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trecem la coord. polare ale unei elipse:

$\begin{cases} x = a \cdot r\cos t \\ y = b \cdot r \sin t \end{cases}$, $r \in [0,1]$, $t \in [0, 2\pi]$, $a = \sqrt{2p}$, $b = \sqrt{\frac{7}{16}}$

$J = a \cdot b \cdot r = \frac{7}{2} r$

$$V = \int_0^{2\pi} \left(\int_0^1 \left(\sqrt{\frac{(a \cdot r\cos t)^2}{64} + (b \cdot r \sin t)^2 + 1} - \sqrt{\frac{7(a \cdot r\cos t)^2}{64} + 7(b \cdot r \sin t)^2 + 1} \right) \cdot \frac{7}{2} r \, dr \right) dt$$

$$= \int_0^{2\pi} \left(\int_0^1 \left(\sqrt{\frac{28 r^2 \cos^2 t}{64_{16}} + \frac{7 r^2 \sin^2 t}{16} + 1} - \sqrt{\frac{7 \cdot 28 r^2 \cos^2 t}{64_{16}} + \frac{7 \cdot 7 r^2 \sin^2 t}{16} + 1} \right) \frac{7}{2} r \, dr \right) dt =$$

$$= \int_0^{2\pi} \left(\int_0^1 \left(\sqrt{\underbrace{\frac{7}{16} r^2 (\cos^2 t + \sin^2 t)}_{=1} + 1} - \sqrt{\underbrace{\frac{49}{16} r^2 (\cos^2 t + \sin^2 t)}_{=1} + 1} \right) \frac{7}{2} r \, dr \right) dt$$

$-\frac{7}{16} + 1 = \frac{9}{16}$

$$= \int_0^{2\pi} \left(\int_0^1 \left(\underbrace{\frac{3}{4} r - \frac{7}{4} r}_{=-r} + 1 \right) \frac{7}{2} r \, dr \right) dt =$$

$$= \int_0^{2\pi} \left(\int_0^1 (1 - r) \frac{7r}{2} \, dr \right) dt =$$

$$= \int_0^{2\pi} \frac{7}{2} \left(\frac{r^2}{2} - \frac{r^3}{3} \right) \Big|_0^1 dt =$$

$$= \int_0^{2\pi} \frac{7}{2} \left(\frac{1}{2} - \frac{1}{3} \right) dt =$$

$$= \frac{7}{2} \cdot \frac{1}{6} t \Big|_0^{2\pi} =$$

$$= \frac{7\pi}{6}$$

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⊛ $f:\{(x,y,z)\in\mathbb{R}^3 \mid 4x^2+9y^2+z^2\le 9\}\to\mathbb{R}$ EXAMEN

$f(x,y,z)=6xy+2xz-x^2-\frac{z^2}{2}$

$\min f=?$, $\max f=?$

$\frac{\partial f}{\partial x}=6y+2z-2x$

$\frac{\partial f}{\partial y}=6x$ $\Rightarrow$ $\begin{cases}6y+2z-2x=0\\6x=0\\2x-z=0\end{cases}$ $\Rightarrow O(0,0,0)$ punct critic $\in$ Int D

$\frac{\partial f}{\partial z}=2x-z$

$Fr\,D=\{(x,y,z)\in\mathbb{R}^3 \mid 4x^2+9y^2+z^2-9=0\}$

fie $Fr\,D=g(x,y,z)=4x^2+9y^2+z^2-9$ „g legătura"

definim fct. lui Lagrange

$F(x,y,z)=f(x,y,z)-\lambda g(x,y,z)$, $\lambda\in\mathbb{R}$ parametru

$F(x,y,z)=6xy+2xz-x^2-\frac{z^2}{2}-\lambda 4x^2-\lambda 9y^2-\lambda z^2+9\lambda$

$\begin{cases}\frac{\partial F}{\partial x}=6y+2z-2x-8\lambda x\\\frac{\partial F}{\partial y}=6x-18\lambda y\\\frac{\partial F}{\partial z}=2x-z-2\lambda z\\g=0\end{cases}$ $\Rightarrow$ $\begin{cases}6y+2z-2x-8\lambda x=0\\6x-18\lambda y=0\\2x-z-2\lambda z=0\\4x^2-9y^2+z^2-9=0\end{cases}$

(2) $\Rightarrow y=\frac{6x}{18\lambda}=\frac{x}{3\lambda}$

(3) $\Rightarrow 2x-z(1+2\lambda)=0\Rightarrow z=\frac{2x}{1+2\lambda}$ $\Big\{$ (1) $\Rightarrow$

$\frac{2x}{\lambda}+\frac{4x}{1+2\lambda}-2x-8\lambda x=0$ $\Rightarrow$ $\frac{2x(1+2\lambda)+4x\cdot\lambda-\lambda(1+2\lambda)(2x+8\lambda x)}{\lambda(1+2\lambda)}=0$

$\Rightarrow$ $\frac{2x+4\lambda x+4\lambda x-(\lambda+2\lambda^2)(2x+8\lambda x)}{\lambda(1+2\lambda)}=0$, $\lambda\neq 0$, $\lambda\neq-\frac{1}{2}$

$\Rightarrow 2x+8\lambda x-2\lambda x-8\lambda^2x-4\lambda^2x-16\lambda^3x=0\Rightarrow$

$\Rightarrow -16\lambda^3x-12\lambda^2x+6\lambda x+2x=0\Rightarrow$

$\Rightarrow x(-16\lambda^3-12\lambda^2+6\lambda+2)=0$

$-8\lambda^3-6\lambda^2+3\lambda+1=0$

observăm $\lambda_1=-1$ soluție

$-8\lambda^3-6\lambda^2+3\lambda+1 : \lambda+1=-8\lambda^2+2\lambda+1$

$+8\lambda^3+8\lambda^2$

$+2\lambda^2+3\lambda+1$

$-2\lambda^2-2\lambda$

$\lambda+1$

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$(\lambda+1)(-8\lambda^2+2\lambda+1)=0$

$-8\lambda^2+2\lambda+1=0$

$\Delta = 4+32=36$

$\lambda_{2,3} = \frac{-2\pm 6}{-16} \Rightarrow \lambda_2 = \frac{1}{2}\ ,\ \lambda_3 = -\frac{1}{4}\ ,\ \lambda_1 = -1$

① $\lambda = -1$

$y = \frac{x}{3\lambda} \Rightarrow y = -\frac{x}{3}$

$z = \frac{2x}{1+2\lambda} = -2x$

$\left.\begin{matrix} \\ \\ \end{matrix}\right\} \overset{(4)}{\Rightarrow} x = \pm\frac{3}{\sqrt{14}}$

$\left(-\frac{3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{6}{\sqrt{14}}\right)\ ,\ \left(\frac{3}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, -\frac{6}{\sqrt{14}}\right)$

② $\lambda = -\frac{1}{4}$

$\left(-\frac{3}{2}, 2, -6\right)\ ,\ \left(\frac{3}{2}, -2, 6\right)$

③ $\lambda = \frac{1}{2}$

$(-3,-2,-3)\ ,\ (3,2,3)$

} puncte critice

$f\left(-\frac{3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{6}{\sqrt{14}}\right) = -\frac{81}{4}$

$f\left(\frac{3}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, -\frac{6}{\sqrt{14}}\right) = -\frac{81}{4}$

$f\left(-\frac{3}{2}, 2, -6\right) = -\frac{81}{4}$

$f\left(\frac{3}{2}, -2, 6\right) = -\frac{81}{4}$

$\left.\begin{matrix} \\ \\ \\ \\ \end{matrix}\right\} \Rightarrow \min$

$f(-3,-2,-3) = 43$

$f(3,2,3) = 43$

$f(0,0,0) = 0$

$\left.\begin{matrix} \\ \\ \\ \end{matrix}\right\} \Rightarrow \max$

extremi GLOBALE

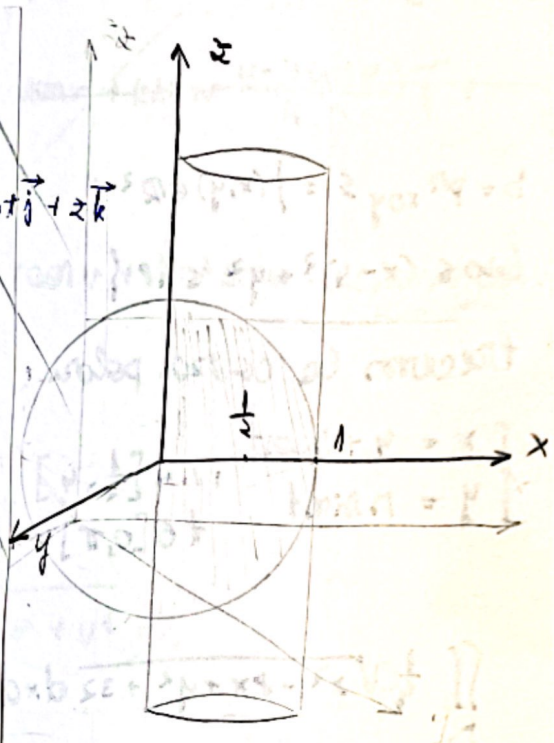
f continuă (fct. elem.)

D domeniu compact (elipsoid) $\Rightarrow$ f este mărginită și își atinge marginile

⊕ Suprafața? POZA 1

$x^2+y^2+z^2=1$ Sferă

$x^2+y^2=x \Rightarrow x^2-x+\frac{1}{4}-\frac{1}{4}+y^2=0 \Rightarrow \left(x-\frac{1}{2}\right)^2+y^2=\frac{1}{4}$ cilindru



parametrizăm cilindrul folosind coord. cilindrice:

$$\begin{cases} x=\frac{1}{2}+\frac{1}{2}\cos t \\ y=\frac{1}{2}\sin t \\ z=z \end{cases} \Rightarrow \vec{r}(z,t)=\left(\frac{1}{2}+\frac{1}{2}\cos t\right)\vec{i}+\frac{1}{2}\sin t\,\vec{j}+z\vec{k}$$

calculăm vectorii tangenți:

$\vec{n}_t=\frac{\partial r}{\partial t}=-\frac{1}{2}\sin t\,\vec{i}+\frac{1}{2}\cos t\,\vec{j}+0\vec{k}$

$\vec{n}_z=\frac{\partial r}{\partial z}=0\vec{i}+0\vec{j}+1\vec{k}$

calculăm vectorul normal:

$$\vec{N}(t,z)=\vec{n}_t\times\vec{n}_z=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\frac{1}{2}\sin t & \frac{1}{2}\cos t & 0 \\ 0 & 0 & 1 \end{vmatrix}=$$

$$=\frac{1}{2}\cos t\,\vec{i}+\frac{1}{2}\sin t\,\vec{j}+0\vec{k}$$

$\vec{N}(t,z)=\left(\frac{1}{2}\cos t,\frac{1}{2}\sin t,0\right)$

calculăm norma vectorului normal:

$\|\vec{N}(t,z)\|=\sqrt{\frac{1}{4}\cos^2 t+\frac{1}{4}\sin^2 t}=\frac{1}{2}$

trecem la calculul integralei:

$$\iint_S d\sigma=\iint_D \|\vec{N}(t,z)\|\,dt\,dz$$

aflăm domeniul D

e bine pe pg. urm.

$x^2+y^2+z^2=1 \Rightarrow z=\sqrt{1-x^2-y^2}$

$x^2+y^2=x$

folosim parametrizarea carteziana:

$$\begin{cases} x=u \\ y=v \\ z=\sqrt{1-u^2-v^2} \end{cases}, (u,v)\in D=\{(u,v)\in\mathbb{R}^2 \mid x^2+y^2\le x\}$$

$$\frac{\partial z}{\partial u}=\frac{-u}{\sqrt{1-u^2-v^2}}, \frac{\partial z}{\partial v}=\frac{-v}{\sqrt{1-u^2-v^2}}$$

$$aria=\iint_S d\sigma=\iint_D \|\vec{N}(u,v)\|\,du\,dv$$

$$\text{prom. cart.} \Rightarrow \vec{N}\left(-\frac{\partial z}{\partial u},-\frac{\partial z}{\partial v},1\right) \Rightarrow \|\vec{N}(u,v)\|=\sqrt{\frac{\partial^2 z}{\partial u^2}+\frac{\partial^2 z}{\partial v^2}+1}$$

$$\Rightarrow aria=\iint_D \sqrt{\frac{u^2}{1-u^2-v^2}+\frac{v^2}{1-u^2-v^2}+1}\,du\,dv=\iint_D \sqrt{\frac{1}{1-u^2-v^2}}\,du\,dv$$

trecem la coordonate polare:

$$\begin{cases} u=r\cos t \\ v=r\sin t \end{cases}, J=r$$

$x^2+y^2\le x \Leftrightarrow r^2\cos^2 t+r^2\sin^2 t\le r\cos t \Leftrightarrow r\le\cos t \Rightarrow r\in[0,\cos t]$

$x^2+y^2\le x \Rightarrow x\ge 0 \Rightarrow \cos t\ge 0 \Rightarrow t\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$$\Rightarrow aria=\int_0^{\pi}\int_0^{\cos t}\frac{r}{\sqrt{1-r^2\cos^2 t-r^2\sin^2 t}}\,dr\,dt=$$

$$=\int_0^{\pi}\int_0^{\cos t}\frac{r}{\sqrt{1-r^2}}\,dr\,dt=\int_0^{\pi}-\frac{1}{2}\int_0^{\cos t}\frac{-2r}{\sqrt{1-r^2}}\,dr\,dt=$$

$$=\int_0^{\pi}-\frac{1}{2}\cdot 2\sqrt{1-r^2}\Big|_0^{\cos t}\,dt=-\int_0^{\pi}\sqrt{1-\cos^2 t}-1\,dt=$$

$$=-\int_0^{\pi}\sin t-1\,dt=\cos t\Big|_0^{\pi}+t\Big|_0^{\pi}=\pi-2$$

(*) $S_1 : z = 2 - \sqrt{x^2+y^2}$ POZA 2

$S_2 : x^2+y^2 = 2x+2y+3$

parametrizăm corbeian : $(x,y,z) \to (u,v,g(u,v))$

$$\begin{cases} x = u \\ y = v \\ z = 2 - \sqrt{u^2+v^2} \end{cases}, \quad D = \{(u,v) \in \mathbb{R}^2 \mid u^2+v^2 \le 2u+2v+3\}$$

$$\text{aria}(S) = \iint_S d\sigma = \iint_D \|\vec{N}(u,v)\| \, du\,dv$$

unde $\vec{N} = \frac{\partial g}{\partial u} \times \frac{\partial g}{\partial v} = \frac{\partial g}{\partial u} i + \frac{\partial g}{\partial v} j + 1k$ și $\|\vec{N}(u,v)\| = \sqrt{\frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial v^2} + 1}$

$$\frac{\partial g}{\partial u} = -\frac{u}{\sqrt{u^2+v^2}}, \quad \frac{\partial g}{\partial v} = -\frac{v}{\sqrt{u^2+v^2}}$$

$$\text{aria}(S) = \iint_D \sqrt{\frac{u^2}{u^2+v^2} + \frac{v^2}{u^2+v^2} + 1}\, du\,dv = \iint_D \sqrt{u^2+v^2}\, du\,dv$$

trecem la coord. polare simple :

$$\begin{cases} u = r\cos t \\ v = r\sin t \end{cases}, \quad J = r$$

$x^2+y^2 = 2x+2y+3 \iff r^2\cos^2 t + r^2\sin^2 t = 2r(\cos t + \sin t) + 3$

$\iff r^2 = 2r(\sin t + \cos t) + 3 \iff r = 2(\sin t + \cos t) + 3$

$r \in [0, 2(\sin t + \cos t) + 3]$, $t \in [0, 2\pi]$?

$$\int_0^{2\pi} \int_0^{a} \sqrt{r^2\cos^2 t + r^2\sin^2 t} \cdot r \, dr\,dt =$$

$$= \int_0^{2\pi} \int_0^{a} r^2 \, dr\,dt = \int_0^{2\pi} \frac{r^3}{3}\Big|_0^{a} dt =$$

$$= \int_0^{2\pi} \frac{(2(\sin t + \cos t) + 3)^3}{3} dt =$$

$$= \frac{1}{3}\int_0^{2\pi} (6\cos t + 6\cos t \sin^2 t + 6\sin t + 6\sin t\cos^2 t + 6 + 3\sin 2t)\, dt$$

$$= \frac{1}{3}\left(6\sin t + \frac{6\sin^3 t}{3} - 6\cos t + \frac{6\cos^3 t}{3} + 6t - 3\cos 2t\right)\Big|_0^{2\pi} =$$

$(\dots) = 4\pi$

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(*) $f: \{(x,y) \in \mathbb{R}^2 / 4(x^2+2x)+3(y^2-4) \le 0\} \to \mathbb{R}$ TEMA 9

$f(x,y) = 4x^2 - 3y^2 - 1$ $\hookrightarrow D$

$\min_{(x,y)\in D} f(x,y) = ?$, $\max_{(x,y)\in D} f(x,y) = ?$

- Int D

$\frac{\partial f}{\partial x} = 8x$

$\frac{\partial f}{\partial y} = -6y$ $\Rightarrow O(0,0) \in D \Rightarrow$ punct critic

- Fr D

aplicăm metoda multiplicatorilor lui Lagrange

fie $F(x,y) = f(x,y) - \lambda g(x,y)$, unde $g(x,y) = 4(x^2+2x)+3(y^2-4)$

$F(x,y) = 4x^2 - 3y^2 - 1 - 4\lambda x^2 - 8\lambda x - 3\lambda y^2 + 12\lambda$

$\frac{\partial F}{\partial x} = 8x - 8x\lambda - 8\lambda$

$\frac{\partial F}{\partial y} = -6y - 6y\lambda$

$g(x,y) = 4x^2 + 8x + 3y^2 - 12$

$\Rightarrow \begin{cases} x - x\lambda - \lambda = 0 \\ -y - y\lambda = 0 \\ 4x^2 + 8x + 3y^2 - 12 = 0 \end{cases}$

(1) $\Rightarrow x(1-\lambda) - \lambda = 0 \Rightarrow x = \frac{\lambda}{1-\lambda}$ (3) $\Rightarrow 4\frac{\lambda^2}{(1-\lambda)^2} + 8\frac{\lambda}{1-\lambda} - 12 = 0$

(2) $\Rightarrow y(-1-\lambda) = 0 \Rightarrow y = 0$

$\Rightarrow \frac{1}{(1-\lambda)^2}\left(4\lambda^2 + 8\lambda(1-\lambda) - 12(1-\lambda)^2\right) = 0$, $\lambda \neq 1$

$4\lambda^2 + 8\lambda - 8\lambda^2 + 24\lambda - 12\lambda^2 = 0$

$-16\lambda^2 + 32\lambda - 12 = 0$

$\lambda_1 = \frac{1}{2} \Rightarrow x_1 = 1 \Rightarrow A(1,0)$

$\lambda_2 = \frac{3}{2} \Rightarrow x_2 = -3 \Rightarrow B(-3,0)$ } puncte critice

f cont. (fcț. elem.)
D mult. compactă (disc) } $\Rightarrow f$ este mărginită și își atinge marginile

$f(0,0) = -1 \Rightarrow$ min

$f(1,0) = 3$

$f(-3,0) = 35 \Rightarrow$ max

$\Rightarrow O(0,0)$ min global

$\Rightarrow B(-3,0)$ max global

(*) Calc. aria $S_1 : x^2+y^2+z^2=4$, $z \geq 0$, care se afla in int. $S_2 : x^2+y^2=2y$

Fie $S_1 \cap S_2 = S$

$\text{Aria}(S) = \iint_S d\sigma$, unde $d\sigma = \|\vec{N}(x,y)\|$

Folosim parametrizarea carteziana, $(x,y,z) \to (u,v,g(u,v))$

$$\begin{cases} x = u \\ y = v \\ z = \sqrt{4-u^2-v^2} \end{cases}$$

Fie $g(u,v) = \sqrt{4-u^2-v^2}$

$$\frac{\partial g}{\partial u} = \frac{-u}{\sqrt{4-u^2-v^2}}, \quad \frac{\partial g}{\partial v} = \frac{-v}{\sqrt{4-u^2-v^2}}$$

$$\vec{N}(u,v) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial g}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial g}{\partial v} \end{vmatrix} = -\frac{\partial g}{\partial u}\vec{i} - \frac{\partial g}{\partial v}\vec{j} + 1\vec{k}$$

$$\|\vec{N}(u,v)\| = \sqrt{\frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial v^2} + 1} = \sqrt{\frac{u^2}{4-u^2-v^2} + \frac{v^2}{4-u^2-v^2} + 1} =$$

$$= \sqrt{\frac{\cancel{u^2}+\cancel{v^2}+4-\cancel{u^2}-\cancel{v^2}}{4-u^2-v^2}} = 2\sqrt{\frac{1}{4-u^2-v^2}}$$

$$\text{Aria}(S) = \iint_D 2\sqrt{\frac{1}{4-u^2-v^2}}\,du\,dv$$

Trecem la coordonate polare simple:

$$\begin{cases} u = r\cos t \\ v = r\sin t \end{cases}, \quad \boxed{|J| = r}$$

$x^2+y^2 \leq 2y \iff r^2 \leq 2r\sin t \Rightarrow r \leq 2\sin t \Rightarrow \boxed{r \in [0, 2\sin t]}$

$x^2+y^2 \leq 2y \iff y \geq 0 \Rightarrow r\sin t \geq 0 \Rightarrow \sin t \geq 0 \Rightarrow \boxed{t \in [0,\pi]}$

Deci, $D = [0, 2\sin t] \times [0,\pi]$

$$\text{Aria}(S) = \int_0^{\pi}\int_0^{2\sin t} \sqrt{\frac{1}{4-r^2\cos^2 t - r^2\sin^2 t}}\,dr\,dt =$$

$$= \int_0^{\pi}\int_0^{2\sin t} 2r\sqrt{\frac{1}{4-r^2}}\,dr\,dt = \int_0^{\pi} -\int_0^{2\sin t} -2r\frac{1}{\sqrt{4-r^2}}\,dr\,dt =$$

$$= \int_0^{\pi} -2\sqrt{4-r^2}\,\Big|_0^{2\sin t}\,dt = \int_0^{\pi}\left(-2\sqrt{4(1-\sin^2 t)} + 2\sqrt{4}\right)dt =$$

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$$= \int_0^{\pi} (-4\cos t + 4)\, dt = -4\sin t + 4t \Big|_0^{\pi} = \boxed{4\pi}$$

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